

The background of the slide is a blurred office environment. It shows several desks with computer monitors, blue ergonomic chairs, and glass partitions. An 'EXIT' sign is visible in the upper left corner. The overall lighting is bright and even.

Modeling for Acquisition Entrepreneurship

AutoResearch Deployed on U.S. Midwest IT MSPs

Yutong Wen | AutoResearch Automates Business Sourcing with Little Costs

AutoResearch Formulation

Sourcing Bottleneck

Acquisition Entrepreneurship looks for "stable but stagnant" unlisted firms. Manual identification requires complex quantitative filtering and deep qualitative due diligence.

Personal Hardship

Historically required exhausting hours of manual data collection and firm-by-firm candidate filtering in the absence of LLMs and data science toolkits.

AUTORESEARCH APPROACH

The Regression Contract

Ordinal regression predicting investment-grade scores (1.0–10.0) in 0.5 steps.

Human-Alignment Audit

Assess agent predictions from the POV of the investment thesis and human judgment.

Automated Optimization

Agent refines the model to fit heuristics → an effective, automated process.

Data, Metric, and Baseline



Sample Scale

90 IT MSP, U.S. Midwest Firms

Sourced from [Apollo.io](https://apollo.io)

70–30 Train–Test–Split

N = 62 Training Firms

(with 5–Fold Cross–Validation)

N = 28 Holdout Test Firms



The Metric

RMSE (Root Mean Squared Error) measures accuracy on our 1–10 scale. Training cross–validation vs. blind holdout.



Success Target

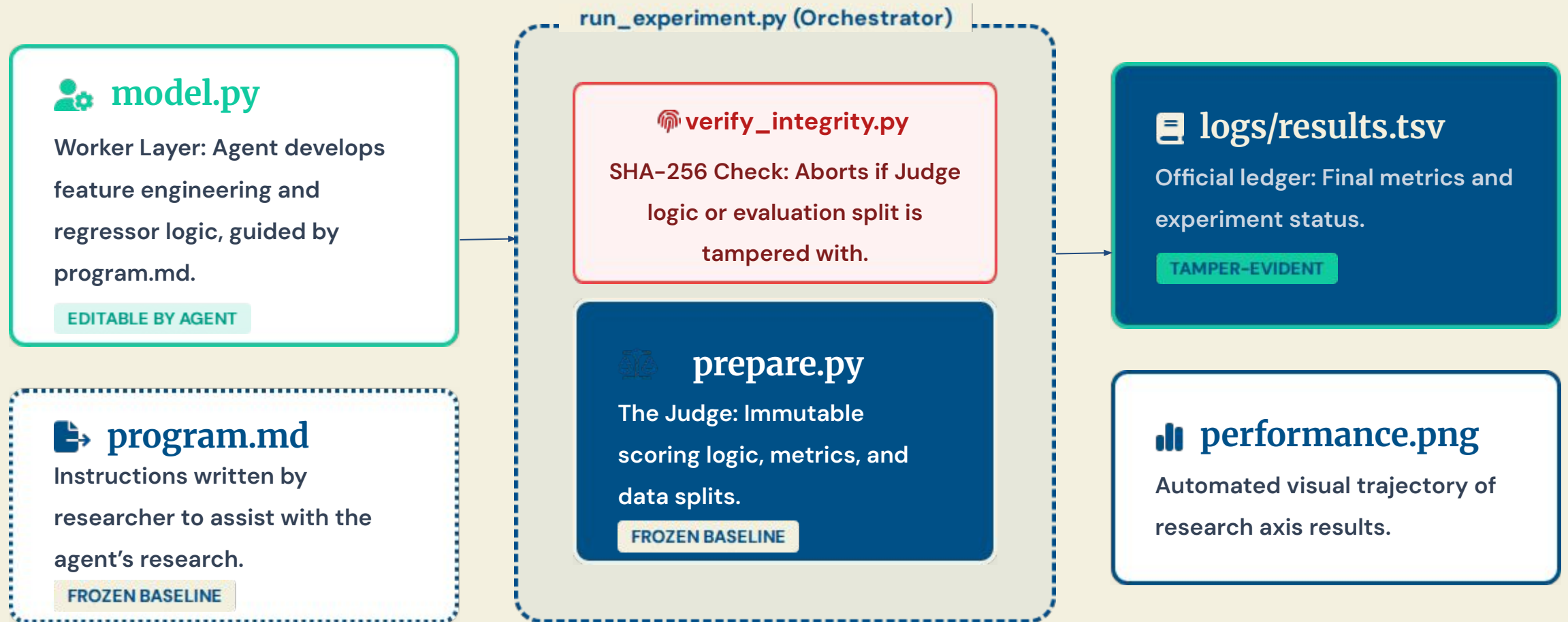
Starting Baseline: 1.846 RMSE

Nested if–else blocks that adds/subtracts points based on tenure and headcounts

Project Success Ceiling:

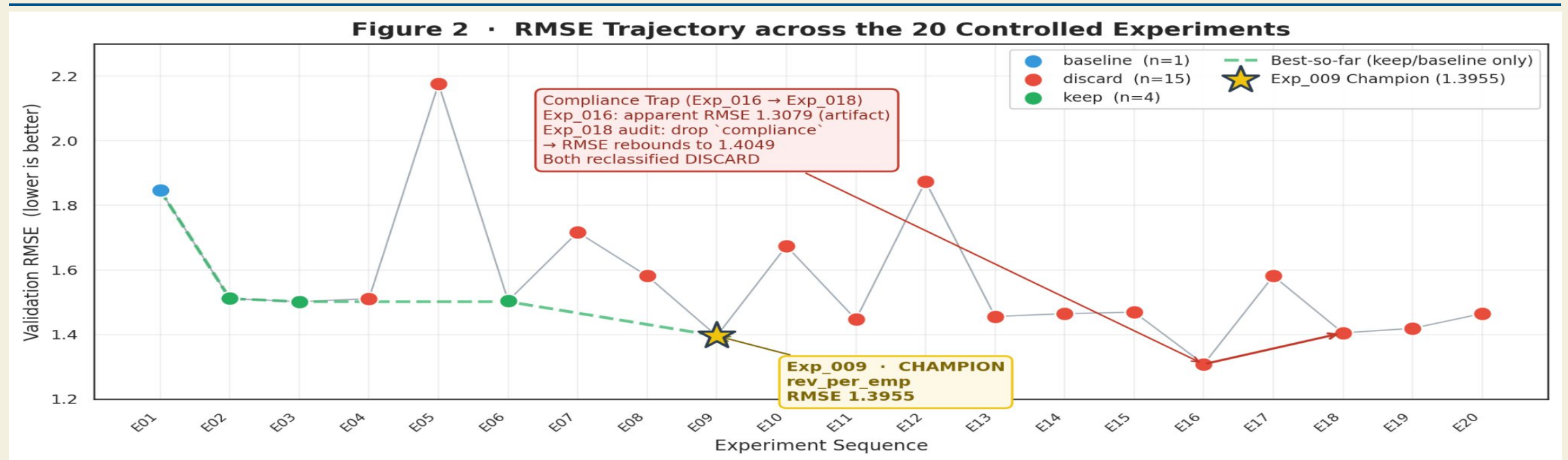
RMSE < 2.0

The AutoResearch Loop Architecture



Key Rule: Evaluation integrity is protected by a SHA-256 One-Way Valve. The agent may freely iterate on *Worker* features, but any modification to the *Judge* logic or evaluation boundaries triggers an immediate system abort.

Experiment Trace: RMSE Trajectory



Significant Turning Points

- **Exp002:** L2-regularized Ridge regression (alpha=1.0) with a quadratic demographic curve (coefficients: tenure +0.66, tenure_sq -0.31)
- **Exp009:** `rev\_per\_emp` to capture the "Stagnation Premium" (operational slack, coefficient: -0.66).

Human Audit - Exp016 & Exp018

- **Exp016 (Illusion):** A binary web-scraped feature for regulatory keywords triggered a drop in RMSE.
- **Trap:** Severe class imbalance. The Ridge model assigned a heavy negative coefficient (-0.58) → unintentional "negation detection" to assign high scores to the 11.3% of firms
- **Exp018 (Ablation Proof):** Stripping the feature from the text parser caused an immediate RMSE rebound.

Final Result: Stability Evidence

Partition	RMSE	R ²	Performance Note
Training (5-fold OOF)	1.3955	0.5128	Champion baseline state.
Holdout Test (N=28)	1.5119	0.4243	Single-pass pass evaluation.
<i>Generalization Gap</i>	+0.1164	-0.0885	8.3% relative inflation (Transferable).

Target < 2.0 achieved with 24.4% safety margin.

Successes and Failures

✓ Worked: Business Heuristics & Structural Facts

- 🏠 Succession Bell Curve (tenure_sq): A non-linear curve that peaks right at the 17–25 year mark captures relationship between company quality and tenure.
- 📊 Stagnation Premium (rev_per_emp): Rewards stable businesses that generate solid revenue despite having low efficiency – perfect reason to acquire it.
- 🔒 SHA-256 Cryptographic Lock: Locked evaluation logic (prepare.py) with a unique cryptographic signature. If the agent tried to alter the test criteria or fabricate better validation scores, the pipeline aborts run.

⚠ Failed: Complex Model on Small Dataset

- 🚫 The 16% Sparsity Floor: Text features that flagged fewer than 16% of the companies introduced extreme sparsity → injected high variance.
- 📄 Bundled Parameters (Exp_012): Forcing multi-variable interactions into the pipeline all at once confused the model's regularizer and crashed.
- 💣 Complex Ensemble Bloat (Exp_011): Heavy multi-model setups (e.g. BaggingRegressor) added unnecessary computational complexity but completely failed to lower RMSE.

Reflection & Loop Limits



Cognitive Split

Agents execute code and log results flawlessly.
Humans remain irreplaceable for auditing semantic flaws and catching data artifacts based on real-world contexts.



Low-Variance Trap

Midwest IT filters limited signal to nearly identical rows. Needs massive cross-industry dataset and let model dynamically learns industry boundaries based on custom investment text prompts.



Extreme Efficiency

Laptop-CPU bound with zero GPU dependency.
Training scales at $O(n^2)$; 10k firms in minutes.

~25s 20
Experiments

\$1.24
Total Cost

N=10k Scaling
Capacity



Human Transition

Role shifts to architecting logic and managing guardrails via **program.md**. Human oversees transparency of experiments while agent maps findings that are readable and understandable by humans.